

DIFFERENTIATED E-LEARNING MODELS BASED ON NEEDS AND ABILITIES OF STUDENTS WITH ARTIFICIAL INTELLIGENCE

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ABSTRACT. *In this study, machine learning is used to solve problems in optimizing performance based on data to classify students based on academic strengths, limitations, and other characteristics. Several artificial intelligence strategies with automatic detection of learning styles in students using machine learning have been presented by various researchers to examine the relationship between academic success and learning preferences. However, no e-learning model currently categorizes students' interests and abilities to promote different learning. This model uses machine learning strategies to evaluate student abilities and demands based on the child's skills and needs. To find out student clusters based on their talents and abilities, artificial intelligence is needed in classifying data. Datasets are grouped to get labels after diagnostic assessment, and then organized by Support Vector Machine (SVM). To separate participant data from groups, SVM will build a hyperplane to determine the style between participants by knowing the type of each participant. Teaching materials and materials will be tailored to each student's interests, talents, and abilities. Based on the test results of 1000 students, the data was effectively separated into three clusters: students in cluster 1 with the trend variable X not knowing and partially understanding, variable Y very ignorant, and Z knew partly with a total of 495 people. Students in cluster 2 with the trend variable X do not know, variable Y does not know and understand part of it, while Z does not know, with a total of 176 people. Students are in cluster 3 with the trend variable X very ignorant, variables Y and Z do not know, with a total of 329 people with an accuracy rate of f1 test score of 100%.*

Keywords: E-learning, Model, Machine learning, K-means clustering

1. Introduction. Developing quality e-learning in an institution requires good standardization [1]. Videos in e-learning learning can be employed as the primary activity in learning and can offer contact with students [2]. Students who engage in self-directed learning actively participate in tasks like reading Internet content [3]. Performance improvement is still a major priority for numerous prestigious business and technology firms. A significant development in computer technology is machine learning, which supports decision-making through prediction algorithms and classification based on data [4]. In recent years, the concept of cloud computing has arisen as a computing paradigm where services are delivered online [5]. The new learning paradigm gives instructors the freedom

to customize learning plans and evaluations to meet the requirements and preferences of individual students.

The goal of the data-driven approach is to create classifiers using data. User models are used in the literature-based approach to learning styles to gather information and produce straightforward if-then rules [6]. When training data is scarce, human and machine learning abilities vary. Performance and the underlying task pattern are highly correlated. While engines perform significantly differently in different ways in experiments [7], humans perform roughly identically across all tasks.

The average length of schooling for Indonesians was less than 2 years in 1950, climbed to 4 years in 1990, and has since quadrupled to 8 years in the present [8]. The decline in the dropout rate contributed to an increase in the gross enrollment rate for junior and senior high schools. GER for Junior High School/Islamic Junior High School equivalent increased from 88.6% in 2014 to 90.6% in 2019. Meanwhile, GER for Senior High School/Vocational High School/Islamic Senior High School equivalent increased from 74.3% in 2014 to 83.98% in 2019.

Diverse initiatives to increase accessibility and raise the standard of education delivery have not yet produced sufficient learning outcomes [9]. However, the expansion of access to education has yet to be fully proportional to the improvement and equity in the quality of education. Regional differences in educational quality continue to be a problem. The results of the last Computer-Based National Examination in 2019 showed that the average score of 2 provinces on the island of Java beat the average score of the highest 10% group in 10 other areas outside Java. Among the things that contribute to the obstacles to increasing and equalizing the quality of education are the competence and performance of teachers. Indonesia's average teacher competency test score is 57 on a scale of 0-100. In addition, teacher creativity in teaching is also an important issue. In comparison to more traditional teaching methods like books and videos, Virtual Reality (VR) is effective at raising student engagement and retention rates for some topics. A student can still be sidetracked and disengaged, though, due to a variety of things, such as unneeded tension, racing thoughts, and noise [10]. The effectiveness of text- and audio-based learning has recently been demonstrated in studies on psychology and human-computer interaction. According to the Modality Principle, on-screen voice is more effective for learning than on-screen text when it comes to intricate graphical representations that need dual-channel working memory processing [11]. Reading texts from digital books outperformed listening to lectures for information retention, according to Baceviciute et al. [12], but there was no discernible difference in knowledge transfer. The results of Han et al.'s [13] research suggested that instruction from real-world teachers may be transferred to online classrooms. Research [14] presented a number of intervention measures to boost students' attentiveness. VR can in some circumstances increase a user's sensation of presence and keep them interested in instructional content [15,16]. In VR, text-based presentations may result in a greater cognitive burden and reduced learning.

Together with the issue of regional differences in educational quality, Indonesia also has a problem with instructors' qualifications. In Indonesia, the teacher-teaching paradigm is still believed to act more as a knowledge transmitter than as a learning facilitator. The capital powering development is stronger when the jobless rate is smaller. The size and makeup of the workforce will keep changing in accordance with current activities in the fields of demography, politics, economics, socioculture, and health. Indonesia is very sensitive to the issue of unemployment. This is indicated by the publication of unemployment data by the Central Bureau of Statistics. Looking at it in detail, the open unemployment rate of Vocational High School graduates is 13.55%, while the lowest is elementary school graduates, which is 3.61%, while the rest, such as Junior High School, at 6.46%, and Senior High School, at 9.86%. Then for diploma, I-III graduates it is 8.08% and for university or strata one graduate it is 7.35%. In August 2020, the Central Bureau of Statistics

reported that there were 9.77 million unemployed persons. This number went up by 2.67 million people from the same period in 2019. Of this number of unemployed, most came from graduates or graduates of Vocational High Schools. The main contributor to the open unemployment rate came from Vocational High School graduates, which amounted to 13.55%.

With various problems in achieving quality resources, especially vocational education, including the disparity in education quality between regions, teachers' competence in Indonesia must be improved. The teacher learning model in Indonesia is believed to be still not on target; the lack of readiness of students to participate in learning, either because of lack of nutrition since childhood, minimal family welfare conditions, as well a lack of basic literacy skills, lack of competence and motivation of teachers in teaching, lack of learning resources, management and governance of education has not developed correctly. Students must teach learning competencies at every stage of primary and secondary education development. The learning outcomes contain several competencies within the scope of the material in the form of a comprehensive narrative, and then adjust the development of learning outcomes in the developmental stages of students who are classified in the age phase [17]. The age phase consists of 6 degrees with synchronization of level, mental age, and chronological age, namely Phase A: Level/grade: Elementary School/Islamic Elementary School (grade 1-2), chronological age: $\leq 6-8$ years, and mental age: ≤ 7 years; Phase B: Level/grade: Elementary School/Islamic Elementary School (grade 3-4), chronological age: 9-10 years, and mental age: ± 8 years; Phase C: Level/grade: Elementary School/Islamic Elementary School (grade 5-6), chronological age: 11-12 years, and mental age: ± 8 years; Phase D: Level/grade: Junior High School/Islamic Junior High School (grade 7-9), chronological age: 13-15 years, and mental age: ± 9 years; Phase E: Level/grade: Senior High School/Islamic Senior High School, Vocational High School/Islamic Vocational High School (grade 10), chronological age: 16-17 years, and mental age: ± 10 years; Phase F: Level/class: Senior High School/Islamic Senior High School, Vocational School/Islamic Vocational High School (grade 11-12), chronological age: 17-23 years, and mental age: ± 10 years. Prior to the teaching and learning process, it is crucial to group pupils according to their interests, talents, and abilities. To create a learning model that is accurate and in line with the demands of students in the workplace and industry, a differentiated e-learning model must be built.

Based on the above problems, namely that no current e-learning model categorizes students' interests and abilities to promote different learning, it is necessary to create an e-learning model that uses machine learning strategies to evaluate students' abilities and demands based on children's skills and needs. Several artificial intelligence strategies with automatic detection of learning styles in students using machine learning have been presented by various researchers to examine the relationship between academic success and learning preference. To solve the problem of optimizing performance based on data to classify students based on academic strengths, limitations, and other characteristics, researchers use machine learning techniques which are part of artificial intelligence. To find out student clusters based on their talents and abilities, artificial intelligence is needed in classifying data. Data sets are clustered for labels after diagnostic assessment, and then managed by a Support Vector Machine (SVM). To separate participant data from groups, SVM will build a hyper plane to determine the style between participants by knowing the type of each participant. Materials and teaching materials will be tailored to each student's interests, talents, and abilities.

2. Material and Methods. By examining models of learning styles, dimensions, values, and combinations to pinpoint students' competencies, strengths, and weaknesses, this research aims to construct a differentiated e-learning model tailored to their requirements and interests. The suggested technique is a machine learning strategy for early detection

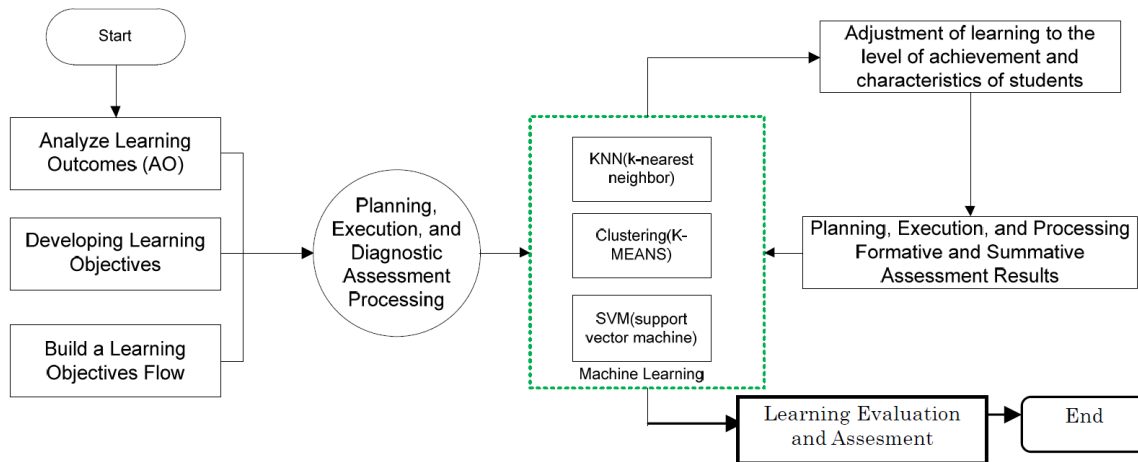


FIGURE 1. Research framework

and the results of analyzing the development of students' learning processes as a research framework using the following concepts.

Learning Outcomes (CP) are learning competencies that students must achieve at each stage of development for each subject in early childhood education units, primary education, and secondary education. Learning outcomes contain a set of competencies and a scope of material comprehensively arranged in narrative form. Adjusting students' developmental stages and mapping learning outcomes is divided into age phases. The purpose of learning achievement analysis activities to develop learning objectives and learning objectives, namely to obtain competency maps that will become a reference for implementing learning. Educators and educational units can use various strategies to develop learning objectives and objectives. It must be ensured that the learning objectives and learning objectives flow and the mapped learning objectives flow meet the criteria.

- a) Competence is the ability that includes attitudes, knowledge, and skills that can be demonstrated by students who show students have succeeded in achieving learning objectives.
- b) Content is core knowledge or main concepts that must be understood at the end of one learning unit.

The development of the flow of learning objectives is carried out at the academic unit level by a team of educators in the same subject or subject group. The flow of learning objectives that have been mapped out is reviewed together within the team, based on the expert judgment of the teaching team, to ensure compliance with the criteria for the flow of learning objectives. Evaluation of the flow of learning objectives should be carried out as part of the overall learning process evaluation. If obstacles or ineffectiveness are identified in learning, improvements can be made to teaching devices, learning objectives, or the flow of learning objectives.

Diagnostic assessment aims to identify students' competencies, strengths, and weaknesses. The teacher uses the results as a reference in planning learning according to the learning needs of students. Under certain conditions, information related to family background, learning readiness, learning motivation, student interests, etc., can be used as material for consideration in planning lessons. To identify the competencies, strengths, and weaknesses of students, a machine learning approach is applied as a basis for initialization and adjustment of learning to the level of achievement and characteristics of students. The input used is to analyze student learning outcomes reports (reports) from the previous year, identify the competencies to be taught, and explore student information on aspects of family background, motivation, interests, learning facilities and infrastructure, and other elements according to the needs of students/schools.

The elements used in planning and achieving optimal e-learning are differentiated.

- 1) Analyze Learning Outcomes (CP), develop Learning Objectives and Learning Objectives Flow
- 2) Diagnostic Assessment Processing
- 3) Development of Teaching Modules
- 4) Adjustment of learning to the level of achievement and characteristics of students
- 5) Planning, Implementation, and Processing of Formative and Summative Assessment Results
- 6) Reporting of Learning Outcomes
- 7) Learning Evaluation and Assessment

The following optimization principles are used to implement the differentiated e-learning model:

- 1) In order to make learning relevant and pleasurable, learning is developed taking account of students' developmental stages and existing academic levels, in accordance with their learning needs, and by reflecting a variety of traits and developments.
- 2) Instruction is planned and carried out to develop learners' capacity to learn continuously.
- 3) The educational process promotes pupils' overall development of competence and character.
- 4) Appropriate learning, or instruction that is planned with parents, the community, and students as partners in accordance with the context, environment, and culture of the pupils.
- 5) Education is focused on a sustainable future.

3. Result and Discussion. Classifying students is conducted to find models or functions to explain or differentiate student data classes to estimate unknown types of students. In general, the classification process has two processes, namely 1) training process: in the training process, a training set is used whose labels are known after a diagnostic assessment is carried out to build a model or function, and 2) testing process: to determine the model's accuracy or procedure built on the training process. In this case, participant data is used, and a testing set is used to predict the labels.

The Support Vector Machine (SVM) is the most commonly used classification algorithm. The most crucial issue in the discussion about classification is the determination of similarity, namely the degree to which the student data resembles the data of other participants. The measure of similarity is essential for clustering [18-21]. The grouping of sets requires understanding the distance for a time series with length w . The simplest option is to treat the subsequence of length w as the element and then apply the Euclidean distance that has been proven empirically. Euclidean distance is a parameter-free method, has fast computation time, and is suitable for various data mining optimizations such as indexing. The meaning is, for $\hat{x} = (x_1, \dots, x_w)$ and $\hat{y} = (y_1, \dots, y_w)$ defined

$$d(\hat{x}, \hat{y}) = \left(\sum_i (x_i - y_i)^2 \right)^{1/2} \quad (1)$$

as a metric in grouping. Other metrics include common metrics that are defined with $L_p(\hat{x}, \hat{y}) = \left(\sum_i (x_i - y_i)^2 \right)^{1/p}$ to $p \geq 1$ and

$$L_\infty = \max_i |x_i - y_i| \quad (2)$$

In various uses, we want to obtain a subsequent shape as the main factor determining the distance. This means that two subsequences can essentially have the same shape even though they have different amplitudes and baselines. One way to achieve this is to

normalize the subsequences and then apply metrics L_2 in normalized subsequences. State the sequence version $\hat{x}$ normalized with $\kappa(\hat{x})$, defined distance between $\hat{x}$ and $\hat{y}$ by

$$d(\hat{x}, \hat{y}) = L_2 (\kappa(\hat{x}) - \kappa(\hat{y})) \quad (3)$$

Normalization can be done in a number of ways $\kappa(\hat{x})_i = x_i - E\hat{x}_i$ (where $E\hat{x}_i$ is the expected or average value of the sequence values), which results in the average sequence value being 0, and can also be used $\kappa(\hat{x})_i = (x_i - E\hat{x})/D\hat{x}$ (where $D\hat{x}$ is the variance of the sequence), which will force the sequence mean to 0 and the variance to 1.

Evaluate the quality of clustering using cross entropy [22] which is stated as follows:

$$Cross\ entropy = \sum_{j=1}^k \left(\frac{n_j}{|SDB|} \right) \left(- \sum_{i=1}^m P_{ij} \log(p_{ij}) \right) \quad (4)$$

k = amount of clusters, n_j = amount of sequences in j cluster, m = amount of classes in sequence database, p_{ij} = probability by random j cluster, SDB = sequence database.

One important comparative metric for assessing cluster or algorithm results is NMI. This can aid academics in analyzing algorithm performance and performance enhancement. For example, C_T and C_E are a set of class labels and a set of cluster labels calculated using the clustering algorithm, and then the intermediate NMI C_T and C_E is

$$NMI(C_T, C_E) = \frac{H(C_T) + H(C_E)}{H(C_T, C_E)} \left(= 1 + \frac{I(C_T; C_E)}{H(C_T, C_E)} \right) \quad (5)$$

where $H(P)$, $H(P, Q)$ and $I(P; Q)$ represent the entropy, joint entropy and mutual information variables P and Q . When C_T and C_E are independent from one variable to another, $NMI(C_T, C_E) = 1$, because $H(C_T, C_E) = H(C_T) + H(C_E)$ must be fulfilled. The greater the $NMI(C_T, C_E)$ the more accurate the cluster results [23]. The selection of sampling density affects sampling efficiency. The cross entropy method can be used to determine a sample density that is almost ideal. The cross entropy method is an adaptive sampling strategy that establishes the sampling density by minimizing the Kullback-Leibler divergence between a predetermined set of parametric distributions and the theoretically ideal required sampling density [24].

The following formulation describes the resulting differentiated e-learning model:

$$ET = \sum_{i=1}^k \sum_{j=1}^l x_{ij} + \frac{X_{ij}}{\sum_{i=1}^n y_{ij}} + \frac{X_{ij}}{\sum_{j=1}^n y_{ij}} + \sqrt{\sum_{k=1}^p (y_{ik} - c_{jk})^2} + \sum_{n=1}^N a_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N a_n a_m t_n t_m k(x_n, x_m) \quad (6)$$

ET = e-learning differentiated, X = matrix element dataset, y = mark element assessment diagnostic, c = distance centroid, N = the number of data labels, a = super vector, t = classification data, k = the number of participants, l = the number of diagnostic assessment variables, i, j, m , and n = index or variable in addition and calculation.

Available data is denoted as x_{ij} , where k is the number of participants and l is the number of diagnostic assessment variables so that $x_{ij} \in k, l$, clusters formed after the diagnostic assessment is denoted by p , so $c \in p$ while the respective labels are denoted $a_n \in \{-1, +1\}$ to $n = 1, 2, \dots, N$, where N is the number of data labels.

Perform diagnostic assessment

$$A = \sum_{i=1}^k \sum_{j=1}^l x_{ij} \quad (7)$$

A = assessment diagnostic, x_{ij} = determine the participant's competencies, strengths, and weaknesses.

$$\sum_{i=1}^k \sum_{j=1}^l x_{ij} > 1, \quad i = 1, \dots, l, \quad j = 1, \dots, k \quad (8)$$

where k = data set or the number of students carrying out a diagnostic assessment while l is the competency variable being tested. Diagnostic assessment data is data from the results of the outcomes of a student's diagnostic evaluation of a specific subject.

1) Normalization using a matrix of unique maximum values

$$P_{ij} = \frac{X_{ij}}{\sum_{i=1}^n y_{ij}} \quad (9)$$

$N^+ = \{(\max P_{ij} | j \in k, i = 1, 2, 3, \dots, n)\}$, $N^- = \{(\min P_{ij} | j \in k, i = 1, 2, 3, \dots, n)\}$, P = diagnostic assessment value.

N^+ is carried out to seek advantage with the potential value of competence, strength, whereas N^- functions to find the value of the cost or constraints to be optimized. By normalizing the matrix, the average and comparison of the diagnostic assessment values between participants were produced.

2) Establish the highest value for each participant.

$$S_{ij} = \frac{X_{ij}}{\sum_{j=1}^n y_{ij}} \quad (10)$$

$NS^+ = \{(\max S_{ij} | j \in k, i = 1, 2, 3, \dots, n)\}$, $NS^- = \{(\min S_{ij} | j \in k, i = 1, 2, 3, \dots, n)\}$.

NS^+ is employed to maximize the value of each participant's diagnostic evaluation, while NS^- to establish the lowest value.

3) Carry out classification and clustering based on talent interests

$$d = \sqrt{\sum_{k=1}^p (x_{ik} - v_{jk})^2} \quad (11)$$

Each component is divided into categories based on its interests, talents, strengths and competences, limitations, and possibly matching restraints. In order to calculate the closeness distance, the distance function d will divide each element into groups based on its shared characteristics.

4) Identify and classify elements with machine learning

$$L(a) = \sum_{n=1}^N a_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N a_n a_m t_n t_m k(x_n, x_m) \quad (12)$$

L = Lagrange.

For the classification of elements that cannot be separated linearly, it must be modified to find a solution, so it must be changed to be more flexible by adding a soft margin hyperplane variable. ξ ($\xi \geq 0, \forall_i : \xi = 0$) if x_i (properly classified) to $x_i \cdot w + b \geq 1 - \xi$ for class 1 and $x_i \cdot w + b \leq 1 - \xi$ for class 2 it will be found to determine the best separator, so the separator field search formula becomes

$$\min \frac{1}{2} |w|^2 + C \left(\sum_{i=1}^n \xi_i \right) \quad (13)$$

$$\text{s.t. } y_i(w \cdot x_i + b) \geq 1 - \xi \quad (14)$$

$$\xi_i \geq 0 \quad (15)$$

C is a parameter to determine how much improvement is due to errors in determining the classification of student data, which is minimized $\frac{1}{2} |w|^2$ equivalent to minimizing

Vapnik-Chervonenkis (VC) which is a property of the hypothetical space and minimizing $C(\sum_{i=1}^n \xi_i)$ error on training data.

It is anticipated that upon classification, two classes – classes -1 and $+1$ – will have emerged. These classes can be completely divided by the value of the d -dimensional hyperplane, which is specified as follows:

$$x_i \cdot w_i + b = 0 \quad (16)$$

Pattern x_i entering the class -1 (negative sample) can be formulated as a pattern that satisfies the inequality:

$$x_i \cdot w_i + b \leq -1 \quad (17)$$

As for pattern x_i which belongs to class $+1$ (positive sample), satisfy the inequality:

$$x_i \cdot w_i + b \leq +1 \quad (18)$$

The largest margin between classes can be found by maximizing the value of the distance between the hyperplane and its closest point, i.e., $1/\|w\|$, can be formulated to find the minimum point with the equation

$$\min \tau(w) = \frac{1}{2} \|w\| \quad (19)$$

$$y_i(x_i \cdot w + b) - 1 \geq 0 \quad (20)$$

For the test, data from Azizi Tanjung Morawa Vocational School, Deli Serdang, were 1000 students who have not used different e-learning in the teaching and learning process, the student datasets are categorized into several different classes using the dividing field equation method. As a result, the learning model is more suited to the students' skills, interests, and talents.

On the basis of similarity, normalized data is categorized. Each element is grouped into three cluster centers using Euclidean distance based on its interests, abilities, strengths, competencies and weaknesses: Cluster Center 1: X: 1.5845, Y: 0.5535, and Z: 0.9375; Center of Cluster 2: X: 0.5282, Y: 1.6605, and Z: 0.4688; Center of Cluster 3: X: 0.5282, Y: 1.1070, and Z: 1.4063. It is successfully classified into 3 clusters, namely 1) Students in grade 1 with the trend variable X do not know and understand some; variable Y does not know, while Z knows some, with a total of 495 people. 2) Students are in grade 2 with the trend variable X very ignorant, and variable Y does not know and partially understands. In contrast, Z understands very much and does not know, with a total of 176 people. 3) Students are in class 3 with the trend variable X very ignorant. Variables Y and Z do not know, with a total of 329.

The results of the normalized cluster dataset are then the training process to produce a classification hyperplane support vector machine to separate the participant data from the class for recommendations for e-learning learning materials according to the style, talents, and interests that are formed. Based on the $k = 1000$ dataset with three parameters, i.e., fully understand, partially understand, and do not know with labels 3, 2, 1. The support vector machine that occurs is a plural class labeled 3, 2, 1. It is included in the plural class because there are more than two classes classified, namely x , y , and z , which are modeled from the results of the diagnostic assessment following the number of fields of the study tested. For instance, the information from the diagnostic assessment of language, social studies, and science was used in this test. After completing the three stages of training, a multi-class SVM classification was created by maximizing the measurement of the separation between the hyperplane and the place where classes are closest. This classification will be used as a knowledge base for data set testing to identify content for Intelligent Tutoring Systems (ITS) that is appropriate for personalizing learning in accordance with the learner's unique characteristics to support knowledge acquisition.

4. Conclusions. The conclusions from this study are as follows. 1) A learning system with a new paradigm that provides flexibility for educators to formulate learning designs and assessments according to the characteristics and needs of students can be optimized by applying artificial intelligence and machine learning. 2) This research produced a differentiated e-learning model that can accommodate all user needs according to abilities, interests, and talents by using artificial intelligence and the application of machine learning to identify the ability level and needs of participants to get a label after a diagnostic assessment. 3) The dataset is clustered to get a label after the diagnostic assessment, and then classified with a support vector machine to separate the participant data from its class. SVM will build a hyperplane whose goal is to know the class between participants by knowing each class. For each participant, the materials and teaching materials will be adapted to the interests, talents, and abilities of each student in Intelligent Tutoring Systems (ITS) which are right on target to personalize learning according to the individual characteristics of students in supporting knowledge acquisition and adaptive learning. 4) Based on test data, it was successfully classified into 3 clusters with an accuracy rate of 100% f1 score testing. That is, students in cluster 1 with the trend variable X do not know and understand part of it; the Y variable does not know, while Z knows partially, with a total of 495 students; students in cluster 2, the trend variable X is very ignorant, and variable Y is innocent and somewhat understanding. In contrast, Z is both very compassionate and clear, with a total of 176; the community and students are in cluster 3 with the tendency of variable X to be very stupid. Y and Z variables do not know, for a total of 329.

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